

Module: Big Data

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Check database & collections

```
> use Bigdata
< already on db Bigdata
> show collections
< bigdb
Bigdata>
```

Create `dept` collection (by inserting data)

Insert Department 10

```
> db.dept.insert({
  _id: 10,
  deptno: 10,
  dname: "ACCOUNTING",
  loc: "NEW YORK"
})
< DeprecationWarning: Collection.insert() is deprecated. Use insertOne, insertMany, or bulkWrite.
< {
  acknowledged: true,
  insertedIds: {
    '0': 10
  }
}
```

```
> db.dept.insert({
  _id: 30,
  deptno: 30,
  dname: "SALES",
  loc: "CHICAGO"
})
< {
  acknowledged: true,
  insertedIds: {
    '0': 30
  }
}
```

2.1

Add Department 30

Insert Multiple Employees

```
> db.emp.insert([
  { empno: 7782, ename: "CLARK", job: "MANAGER", sal: 2450, deptno: 10 },
  { empno: 7839, ename: "KING", job: "PRESIDENT", sal: 5000, deptno: 10 }
])
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69bb82a679cfa3a8c5bee1b9'),
    '1': ObjectId('69bb82a679cfa3a8c5bee1ba')
  }
}
```

Insert Employees for Department 30

```
> db.emp.insert([
  {
    empno: 7499,
    ename: "ALLEN",
    job: "SALESMAN",
    mgr: 7698,
    hiredate: new Date("1995-02-20"),
    sal: 1600,
    comm: 300,
    deptno: 30
  },
  {
    empno: 7698,
    ename: "BLAKE",
    job: "MANAGER",
    mgr: 7839,
    hiredate: new Date("1981-05-01"),
    sal: 2850,
    deptno: 30
  },
  {
    empno: 7900,
    ename: "JAMES",
    job: "CLERK",
    mgr: 7698,
    hiredate: new Date("1981-12-03"),
    sal: 950,
    deptno: 30
  },
  {
    empno: 7654,
    ename: "MARTIN",
    job: "SALESMAN",
    mgr: 7698,
    hiredate: new Date("1993-09-28"),
    sal: 1250,
    comm: 1400,
    deptno: 30
  },
  {
```

```
    empno: 7844,  
    ename: "TURNER",  
    job: "SALESMAN",  
    mgr: 7698,  
    hiredate: new Date("1981-09-08"),  
    sal: 1500,  
    comm: 0,  
    deptno: 30  
  },  
  {  
    empno: 7521,  
    ename: "WARD",  
    job: "SALESMAN",  
    mgr: 7698,  
    hiredate: new Date("1994-02-22"),  
    sal: 1250,  
    comm: 500,  
    deptno: 30  
  }  
])  
< {  
  acknowledged: true,  
  insertedIds: {  
    '0': ObjectId('69bb82ba79cfa3a8c5bee1bb'),  
    '1': ObjectId('69bb82ba79cfa3a8c5bee1bc'),  
    '2': ObjectId('69bb82ba79cfa3a8c5bee1bd'),  
    '3': ObjectId('69bb82ba79cfa3a8c5bee1be'),  
    '4': ObjectId('69bb82ba79cfa3a8c5bee1bf'),  
    '5': ObjectId('69bb82ba79cfa3a8c5bee1c0')  
  }  
}
```

Querying Data

Find all

```
> db.emp.find()
< {
  _id: ObjectId('69bb82a679cfa3a8c5bee1b9'),
  empno: 7782,
  ename: 'CLARK',
  job: 'MANAGER',
  sal: 2450,
  deptno: 10
}
{
  _id: ObjectId('69bb82a679cfa3a8c5bee1ba'),
  empno: 7839,
  ename: 'KING',
  job: 'PRESIDENT',
  sal: 5000,
  deptno: 10
}
{
  _id: ObjectId('69bb82ba79cfa3a8c5bee1bb'),
  empno: 7499,
  ename: 'ALLEN',
  job: 'SALESMAN',
  mgr: 7698,
  hiredate: 1995-02-20T00:00:00.000Z,
  sal: 1600,
  comm: 300,
  deptno: 30
}
{
  _id: ObjectId('69bb82ba79cfa3a8c5bee1bc'),
  empno: 7698,
  ename: 'BLAKE',
  job: 'MANAGER',
  mgr: 7839,
  hiredate: 1981-05-01T00:00:00.000Z,
  sal: 2850,
  deptno: 30
}
{
  _id: ObjectId('69bb82ba79cfa3a8c5bee1bd'),
```

```
    job: 'CLERK',
    mgr: 7698,
    hiredate: 1981-12-03T00:00:00.000Z,
    sal: 950,
    deptno: 30
  }
  {
    _id: ObjectId('69bb82ba79cfa3a8c5bee1be'),
    empno: 7654,
    ename: 'MARTIN',
    job: 'SALESMAN',
    mgr: 7698,
    hiredate: 1993-09-28T00:00:00.000Z,
    sal: 1250,
    comm: 1400,
    deptno: 30
  }
  {
    _id: ObjectId('69bb82ba79cfa3a8c5bee1bf'),
    empno: 7844,
    ename: 'TURNER',
    job: 'SALESMAN',
    mgr: 7698,
    hiredate: 1981-09-08T00:00:00.000Z,
    sal: 1500,
    comm: 0,
    deptno: 30
  }
  {
    _id: ObjectId('69bb82ba79cfa3a8c5bee1c0'),
    empno: 7521,
    ename: 'WARD',
    job: 'SALESMAN',
    mgr: 7698,
    hiredate: 1994-02-22T00:00:00.000Z,
    sal: 1250,
    comm: 500,
    deptno: 30
  }
}
```

Pretty format

The data comes back messy. The `pretty()` function can be used to improve the layout:

```
> db.emp.find().pretty()
< {
  _id: ObjectId('69bb82a679cfa3a8c5bee1b9'),
  empno: 7782,
  ename: 'CLARK',
  job: 'MANAGER',
  sal: 2450,
  deptno: 10
}
{
  _id: ObjectId('69bb82a679cfa3a8c5bee1ba'),
  empno: 7839,
  ename: 'KING',
  job: 'PRESIDENT',
  sal: 5000,
  deptno: 10
}
{
  _id: ObjectId('69bb82ba79cfa3a8c5bee1bb'),
  empno: 7499,
  ename: 'ALLEN',
  job: 'SALESMAN',
  mgr: 7698,
  hiredate: 1995-02-20T00:00:00.000Z,
  sal: 1600,
  comm: 300,
  deptno: 30
}
{
  _id: ObjectId('69bb82ba79cfa3a8c5bee1bc'),
  empno: 7698,
  ename: 'BLAKE',
  job: 'MANAGER',
  mgr: 7839,
  hiredate: 1981-05-01T00:00:00.000Z,
  sal: 2850,
  deptno: 30
}
{
  _id: ObjectId('69bb82ba79cfa3a8c5bee1bd'),
```

With condition CLERK

```
> db.emp.find({job: "CLERK"})
< {
  _id: ObjectId('69bb82ba79cfa3a8c5bee1bd'),
  empno: 7900,
  ename: 'JAMES',
  job: 'CLERK',
  mgr: 7698,
  hiredate: 1981-12-03T00:00:00.000Z,
  sal: 950,
  deptno: 30
}
```

Greater than

```
> db.emp.find({sal: {$gt: 2400}})
< {
  _id: ObjectId('69bb82a679cfa3a8c5bee1b9'),
  empno: 7782,
  ename: 'CLARK',
  job: 'MANAGER',
  sal: 2450,
  deptno: 10
}
{
  _id: ObjectId('69bb82a679cfa3a8c5bee1ba'),
  empno: 7839,
  ename: 'KING',
  job: 'PRESIDENT',
  sal: 5000,
  deptno: 10
}
{
  _id: ObjectId('69bb82ba79cfa3a8c5bee1bc'),
  empno: 7698,
  ename: 'BLAKE',
  job: 'MANAGER',
  mgr: 7839,
  hiredate: 1981-05-01T00:00:00.000Z,
  sal: 2850,
  deptno: 30
}
```

Updating Data

Update one

```
> db.emp.update(  
  {empno: 7782},  
  {$set: {job: "ANALYST"}}  
)  
< DeprecationWarning: Collection.update() is deprecated. Use updateOne, updateMany, or bulkWrite.  
< {  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 1,  
  modifiedCount: 1,  
  upsertedCount: 0  
}
```

Increase salary

```
> db.emp.updateMany({}, {$inc: {sal: 100}})  
< {  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 8,  
  modifiedCount: 8,  
  upsertedCount: 0  
}
```

Exercise 2.4

Update department 40 name

```
> db.dept.update(  
  {deptno: 40},  
  {$set: {dname: "COMPUTING"}}  
)  
< {  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 0,  
  modifiedCount: 0,  
  upsertedCount: 0  
}
```

Update salary of employee 7788

```
> db.emp.update(  
  {empno: 7788},  
  {$set: {sal: 3500}}  
)  
< {  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 0,  
  modifiedCount: 0,  
  upsertedCount: 0  
}
```

Deleting Data

Delete one

```
> db.dept.deleteOne({deptno: 40})
< {
  acknowledged: true,
  deletedCount: 0
}
```

Delete all matching

```
> db.dept.remove({deptno: 40})
< DeprecationWarning: Collection.remove() is deprecated. Use deleteOne, deleteMany, findOneAndDelete, or bulkWrite.
< {
  acknowledged: true,
  deletedCount: 0
}
```

Aggregation

Filter employees with salary > 2000

```
> db.deptCollection.aggregate([
  {
    $project: {
      empSet: {
        $filter: {
          input: "$employees",
          as: "employee",
          cond: { $gte: ["$$employee.sal", 2000] }
        }
      }
    }
  }
])
<
Bigdata >
```